

NATURE OF STATISTICS

Anthony S. Morfe_CAS_Math Cluster

OBJECTIVES:

At the end of this presentation, students should be able to

- Define basic concepts of statistics.
- Distinguish samples and populations.
- Identify types of data and the measurement level for each variable.

A branch of mathematics that deals with the scientific collection, organization, presentation, analysis, and interpretation of numerical data in order to obtain useful and meaningful information. (General)



DEFINITION OF STATISTICS

A set of procedures and rules for reducing large masses of data into manageable proportions allowing us to draw conclusions from those data. (McCarthy)



DEFINITION OF STATISTICS

A person who is trained in collecting numerical information (data), evaluating it, and drawing conclusions from it.

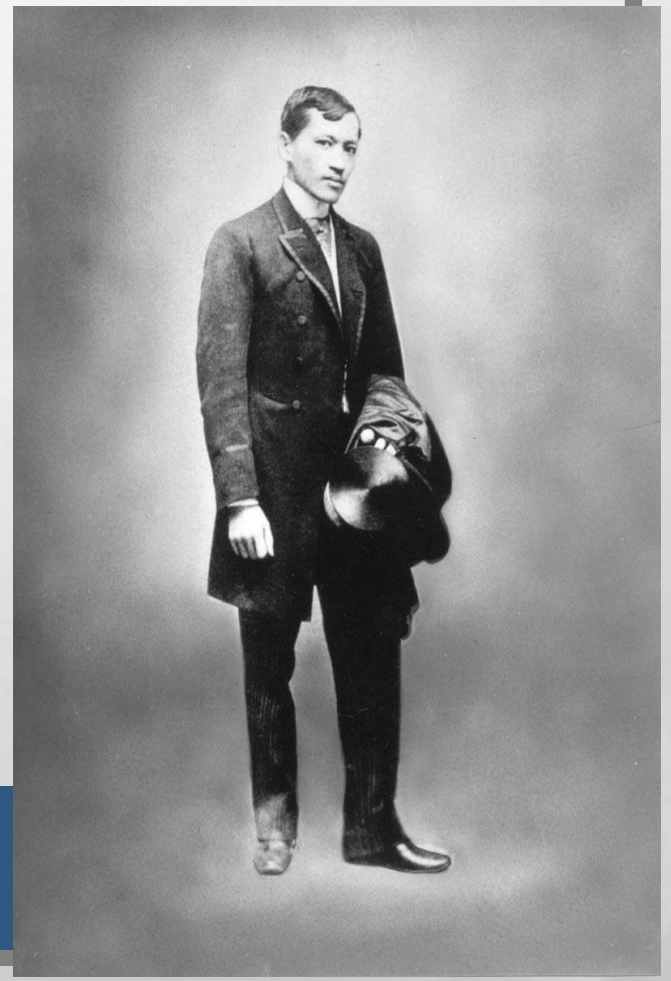


STATISTICIAN

- **Measurement** – assignment of a number to something.

Example:

The height of Gat. Jose Rizal is 4'11".



BASIC CONCEPTS

- Variable (data) - characteristics or attribute that can assume different values

  Examples:

  > allowance of MCL students

  > opinion of the students about federalism



BASIC CONCEPTS

Data Values- values (measurements or observations) that the variables can assume.

Example: (Scores in a 40-item quiz)

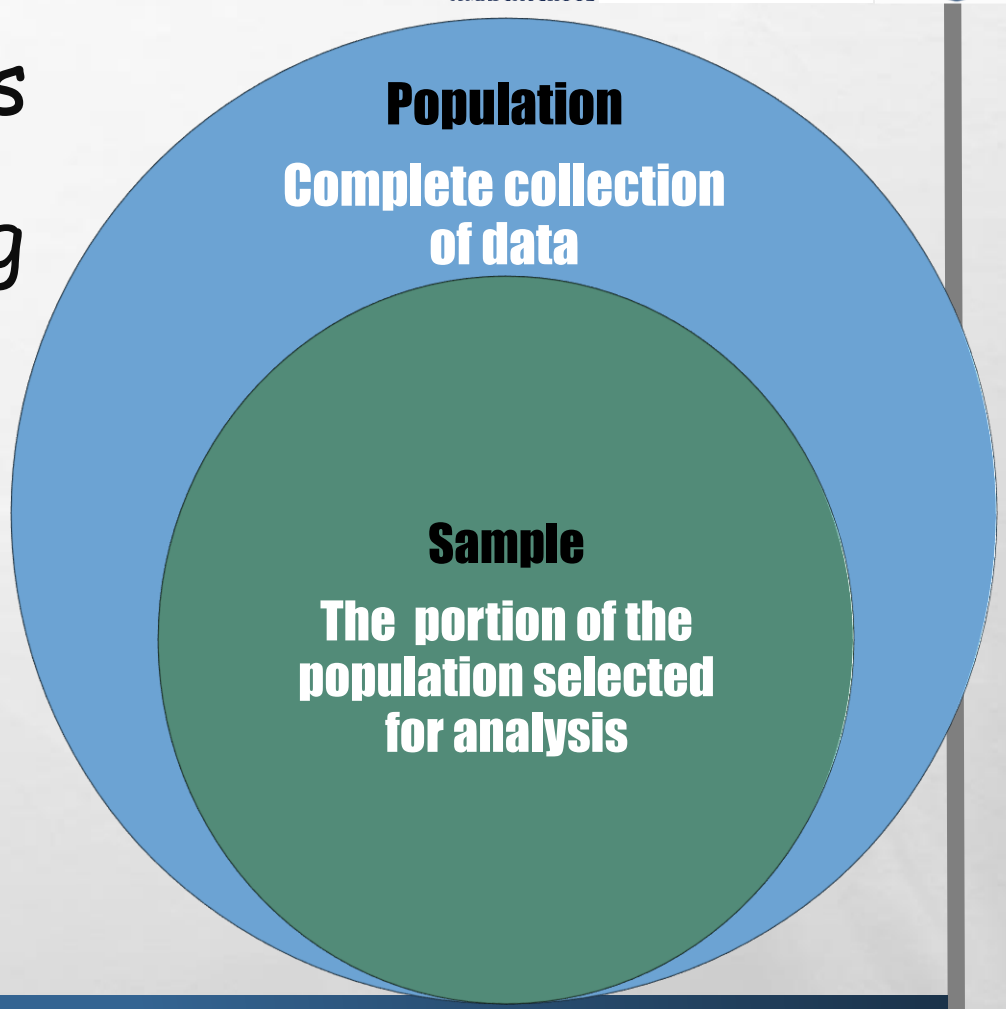
21,27,19,21,20,24,30



BASIC CONCEPTS

- *A population* consists of all subjects (human or otherwise) that are being studied.

- *A sample* is a group of subjects selected from a population.

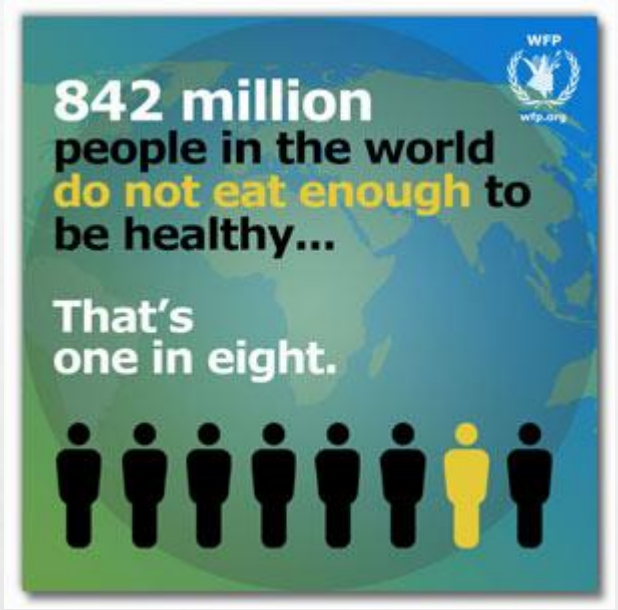


BASIC CONCEPTS

Measure used to describe the population is called the PARAMETER.

Measure computed from sample data is called STATISTIC.

Population vs. Sample



TYPES OF STATISTICS

- Descriptive statistics consists of the collection, organization, summarization, and presentation of data.

  Example:

  Describing the allocation of your weekly allowance

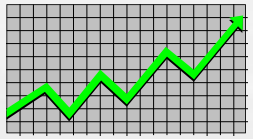
TYPES OF STATISTICS

DESCRIPTIVE STATISTICS

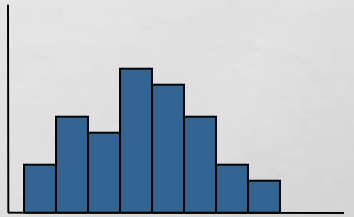
- COLLECT DATA
  - SURVEY



- PRESENT DATA
  - TABLES AND GRAPHS

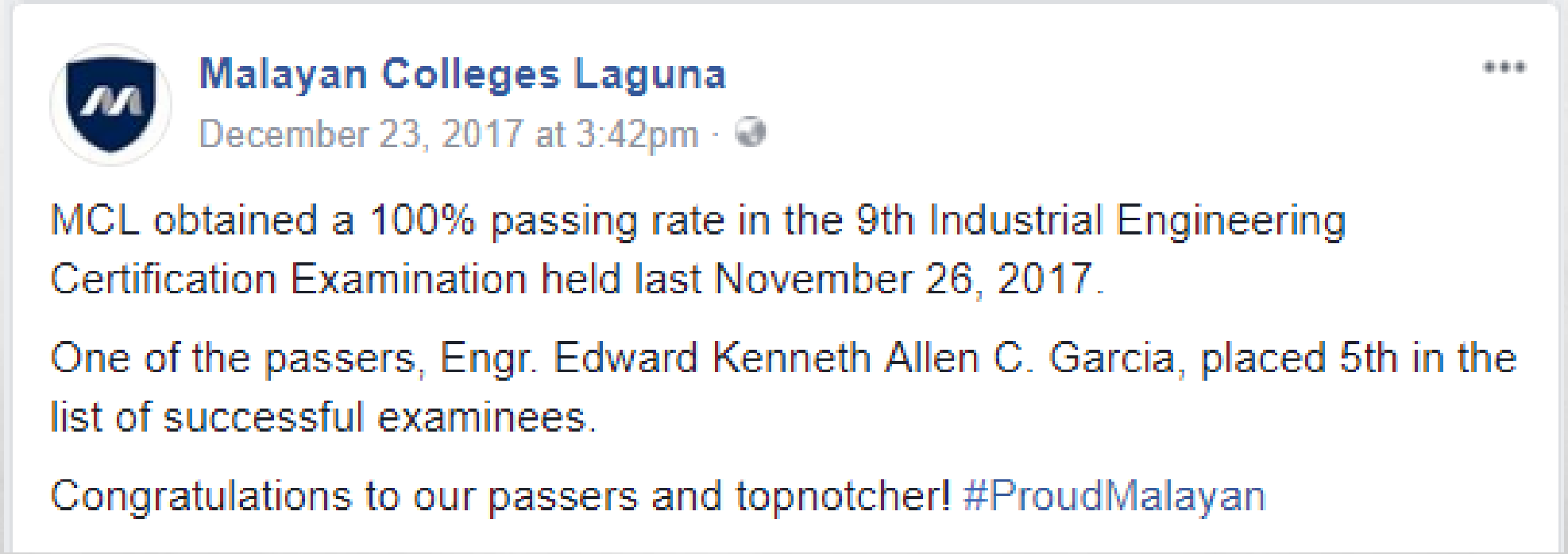


- CHARACTERIZE DATA
  - SAMPLE MEAN = $\frac{\sum X_i}{n}$



DESCRIPTIVE STATISTICS

EXAMPLE:



Malayan Colleges Laguna
December 23, 2017 at 3:42pm ·

MCL obtained a 100% passing rate in the 9th Industrial Engineering Certification Examination held last November 26, 2017.

One of the passers, Engr. Edward Kenneth Allen C. Garcia, placed 5th in the list of successful examinees.

Congratulations to our passers and topnotcher! #ProudMalayan

DESCRIPTIVE STATISTICS

EXAMPLE:



Source:
http://espn.go.com/nba/player/stats/_/id/3975/stephen-curry

- Inferential statistics consists of generalizing from samples to populations, performing estimations and hypothesis tests, determining relationships among variables, and making predictions.

TYPES OF STATISTICS

INFERENTIAL STATISTICS

- **ESTIMATION AND HYPOTHESIS TESTING**
  - estimate the population mean weight using the sample mean weight
  - test the claim that the population mean weight is 120 pounds



INFERENTIAL STATISTICS



SEVERAL WELL-DESIGNED LONG-TERM CLINICAL STUDIES HAVE FOUND THAT PEOPLE WHO TAKE BLOOD-PRESSURE-LOWERING DRUGS ACTUALLY SUFFER FROM UNNECESSARY SIDE EFFECTS INCLUDING AN INCREASED RISK OF HEART DISEASE.

SOURCE: HTTP://WHFOODS.ORG/GENPAGE.PHP?TNAME=DISEASE&DBID=15

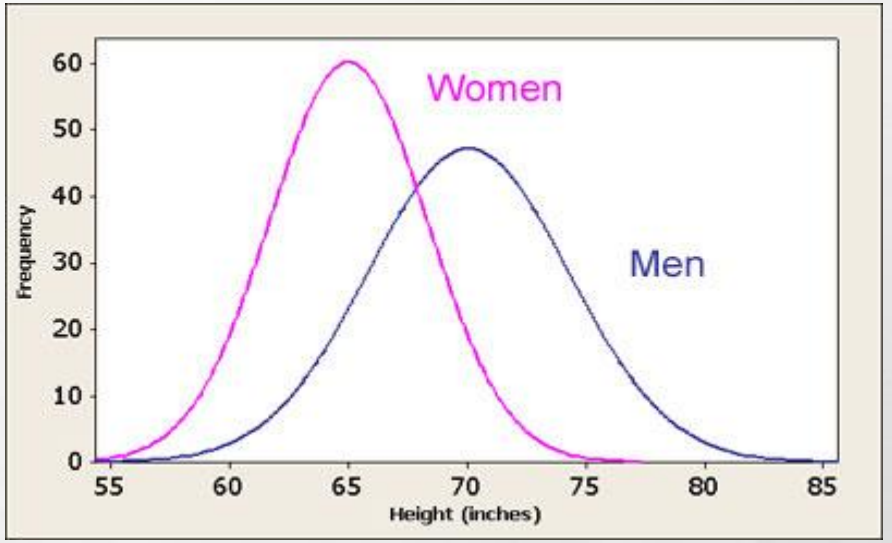
TYPES OF STATISTICS

- Inferential statistics

Hypothesis Testing:

Is there a significant difference between the height of male and female MCL students?

Decision: There is a significant difference between the height of male and female MCL students.





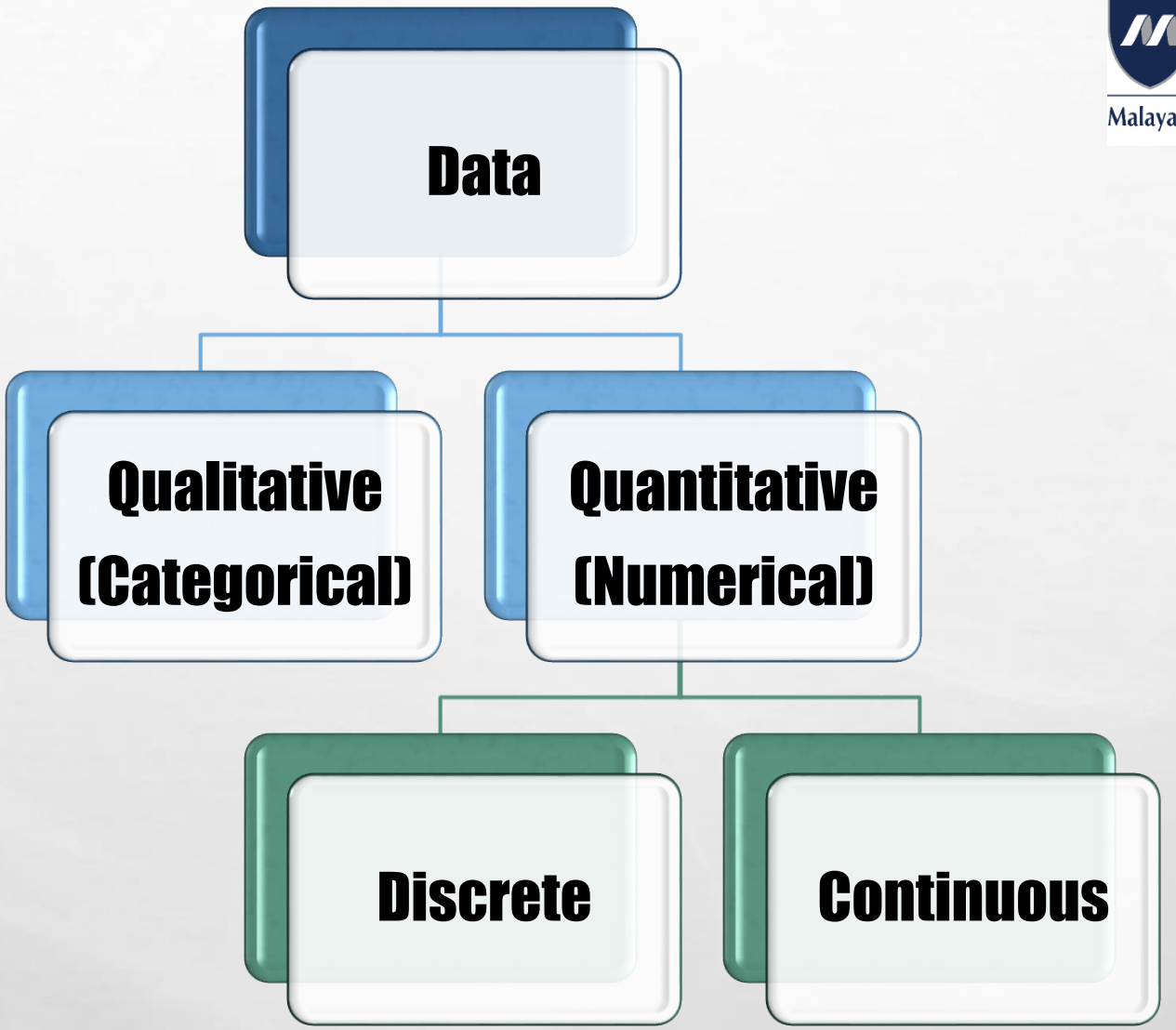
- In each of these statements, tell whether descriptive or inferential statistics have been used.

1. On average, 100 people choke to death on ball point pens every year. (statisticbrain.com) **1. Descriptive**
2. The average person's left hand does 56% of the typing. (statisticbrain.com) **2. Descriptive**
3. By 2040 at least 3.5 billion people will run short of water (World Future Society). **3. Inferential**
4. Allergy therapy makes bees go away (Source: Prevention) **4. Inferential**
5. Nerve impulses to and from the brain travel as fast as 170 miles per hour. (typepad.com) **5. Descriptive**

LET'S PRACTICE

TYPES OF DATA





TYPES OF DATA

- Qualitative data are variables that can be placed into distinct categories, according to some characteristic or attribute.

✓ consist of labels, category names, and such for which representation on a numerical scale is not naturally meaningful

Examples:

❖ Opinion of Catholics to Death Penalty (Pro or Anti)

❖ Name of your friends in MCL

TYPES OF DATA

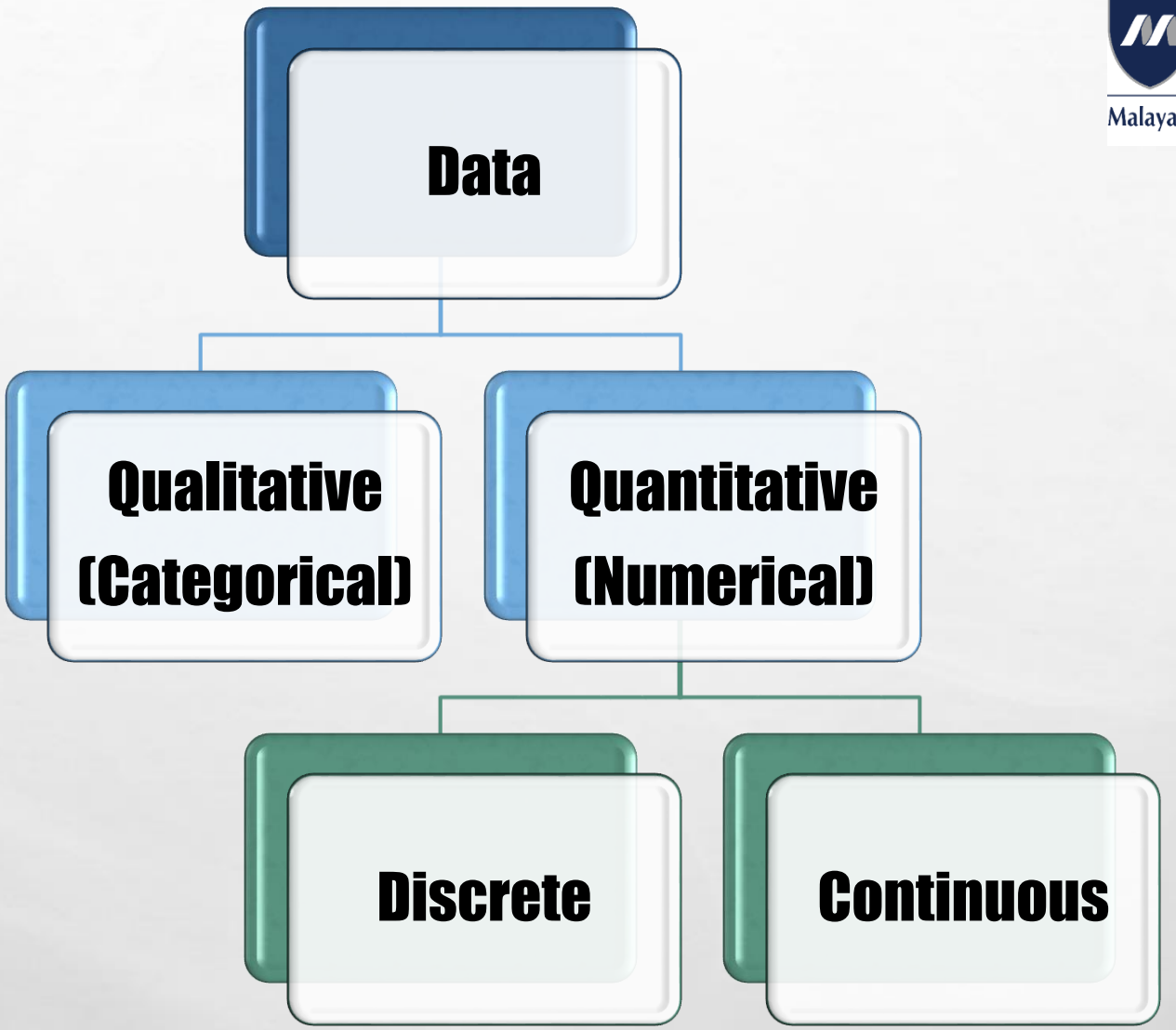
- Quantitative data are numerical and can be ordered or ranked.

✓ are counts or measurements for which representation on a numerical scale is naturally meaningful.

Example:

Amount of a student's daily allowance.

TYPES OF DATA



TYPES OF DATA

Discrete Data

- quantitative data that are countable using a finite count, such as 0, 1, 2, and so on
- integer-valued

Continuous Data

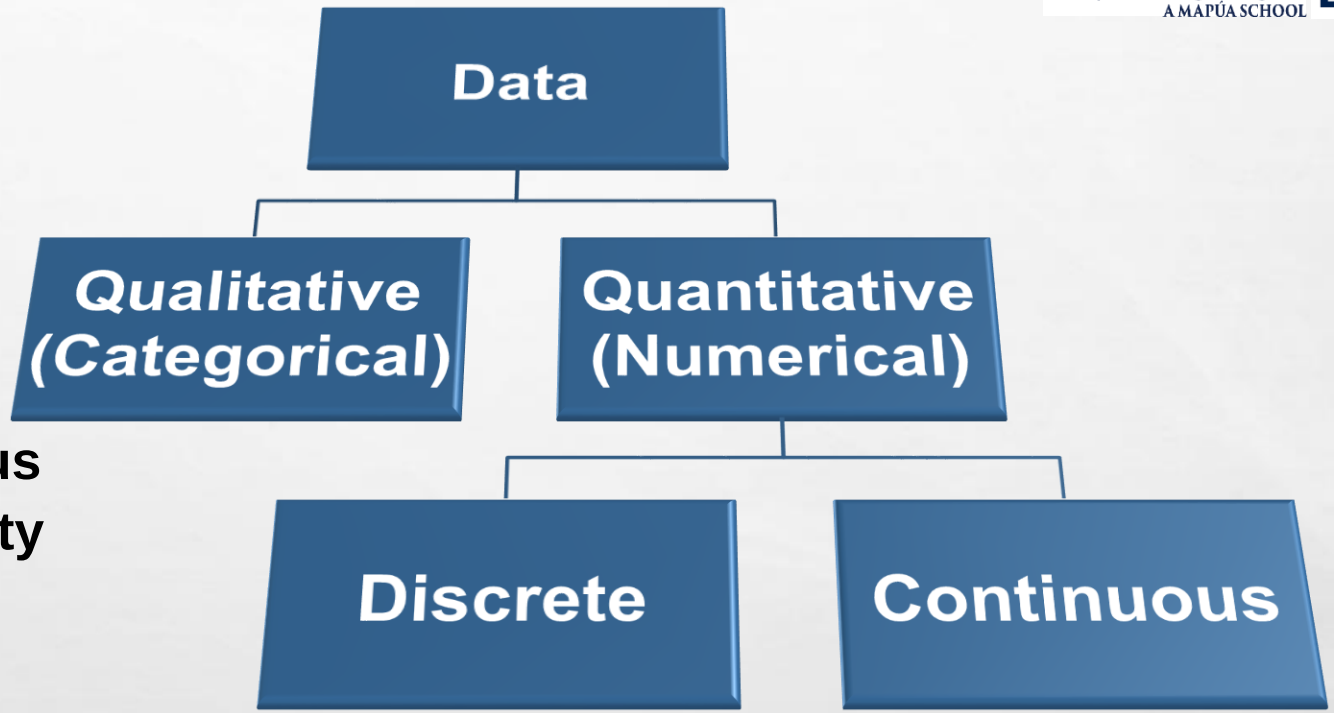
- quantitative data that can take on any value within a range of values on a numerical scale in such a way that there are no gaps, jumps, or other interruptions
- real-valued

DISCRETE OR CONTINUOUS?

Examples

- Daytime temperature readings (in degrees Fahrenheit) in a 30-day period **continuous**
- Ages of MATH111 students **continuous**
- Number (0, 1, 2, or so on) of people attending a conference **discrete**
- Defects per hour in a shoe company **discrete**
- Number of hours you waited for your girlfriend **continuous**

DATA TYPES



Examples:

- Marital Status
- Political Party
- Eye Color

(Defined categories)

Examples:

- Number of Children
- Defects per hour

(Counted items)

Examples:

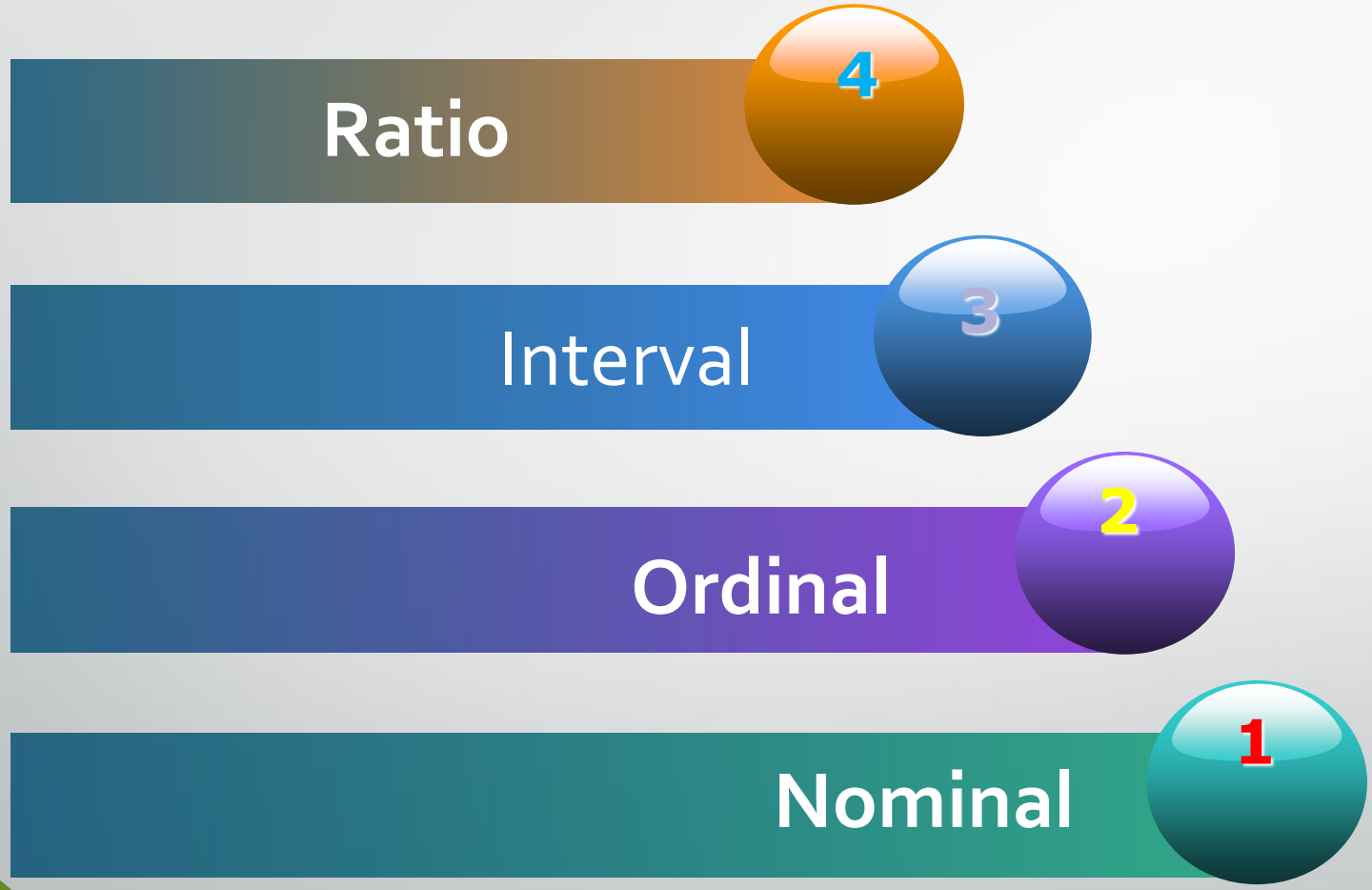
- Weight
- Voltage
- Sales

(Measured characteristics)

LEVELS OF MEASUREMENT

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Levels of Measurement



Nominal Scale

- the lowest level of data
- applied to data that are used for category identification
- characterized by data that consist of names, labels, or categories only
- data cannot be arranged in an ordering scheme
    - arithmetic operations are not performed for nominal data

Nominal Scale

Qualitative Variable	Data Values
Blood type	A, B, AB, O
Gender	male, female
Status	single, married, separated
Name of Schools	MCL, MIT, UP, ADMU, DLSU

Nominal Scale

Qualitative variable	Possible nominal level data values
Province of residence	Laguna, Batangas, Cavite, Rizal, Quezon
Color of road signs	red, white, blue, green
Religion	Christian, Catholic, etc.

Ordinal Scale

- the next higher level of data
- characterized by data that applies to categories that can be ranked
  - data can be arranged in an ordering scheme
  - arithmetic operations are not performed on ordinal level data

Ordinal Scale

Qualitative variable	Data values
Product rating	Poor, good, excellent
Socioeconomic class	Lower, middle, upper
Pain level	None, low, moderate, severe

Example Likert Scale

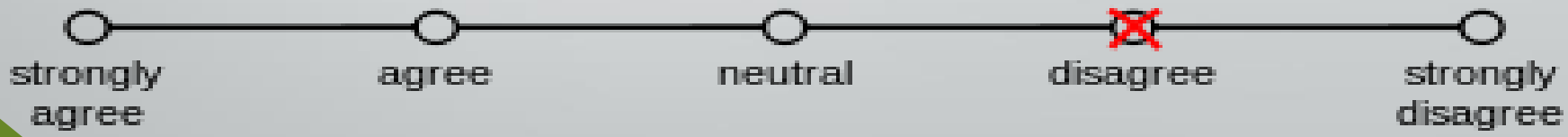
1. Wikipedia has a user friendly interface.



2. Wikipedia is usually my first resource for research.



3. Wikipedia pages generally have good images.



Interval Scale

- applied to data that can be arranged in some order and for which differences in data values are meaningful
- results from counting or measuring
- the value zero is arbitrarily chosen for interval data and does not imply an absence of the characteristic being measured
    - Ex: temperature, IQ scores

Ratio Scale

- the highest level of measurement
- applied to data that can be ranked and for which all arithmetic operations including division can be performed
- results from counting or measuring
- data can be arranged in an ordering scheme and differences and ratios can be calculated and interpreted

Ratio Scale

- data has an absolute zero and a value of zero indicates a complete absence of the characteristic of interest
- Examples:

  wages height weight

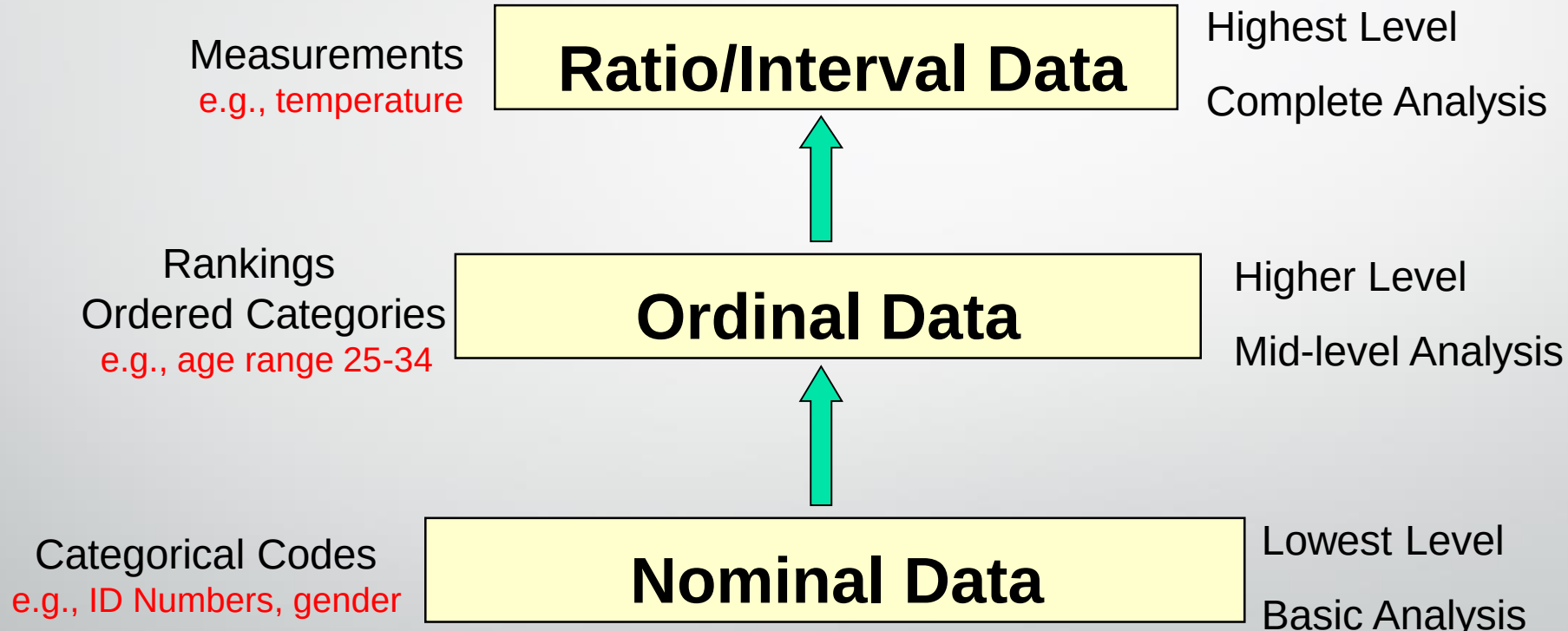
  units of production

  changes in stock prices

  distance between branch offices

  grams of fats consumed per day

Data Measurement Levels



SAMPLING METHODS

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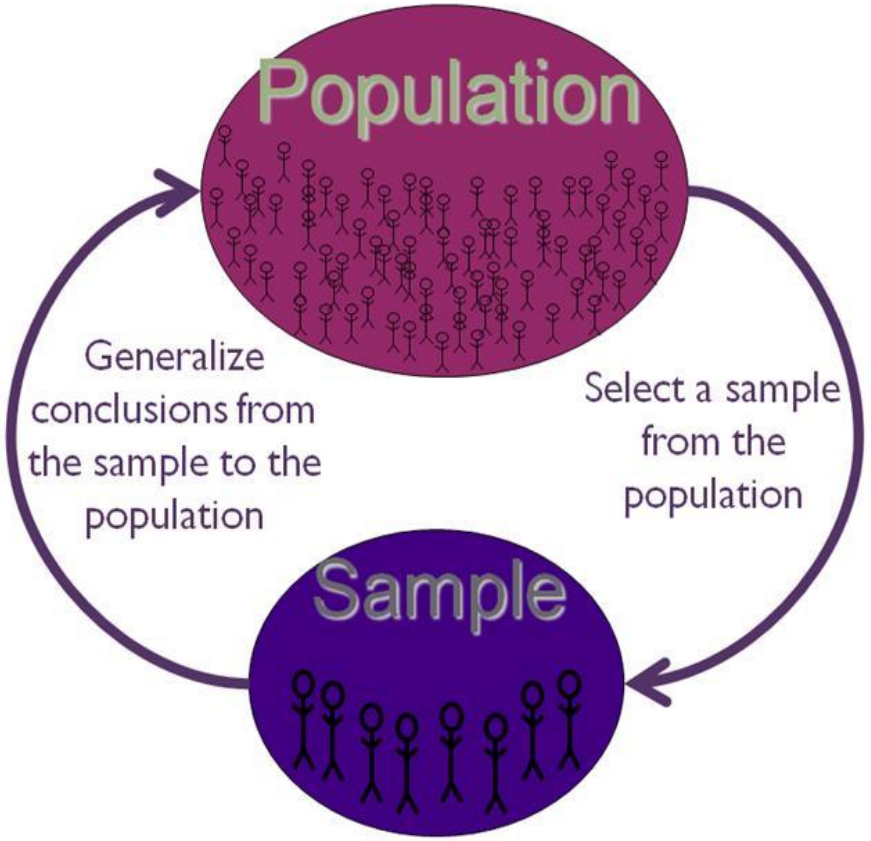
- **Population**. All of the subjects of interest.
- **Sample**. The subjects in the population we actually measure.
- **Sampling**. The process of selecting the individuals from the population that makes up our sample.

The details of which subjects are and are not part of our population should be carefully specified.

- our sample is our only source of information about the population

The **theory of sampling** is as follows:

- Researchers want to gather information about a whole group of people (the **population**).
- Researchers can only observe a part of the population (the **sample**).
- The findings from the sample are **generalized**, or extended, back to the population.



Sampling

- Less time consuming than a census
- Less costly to administer than a census
- It is possible to obtain statistical results of a sufficiently high precision based on samples

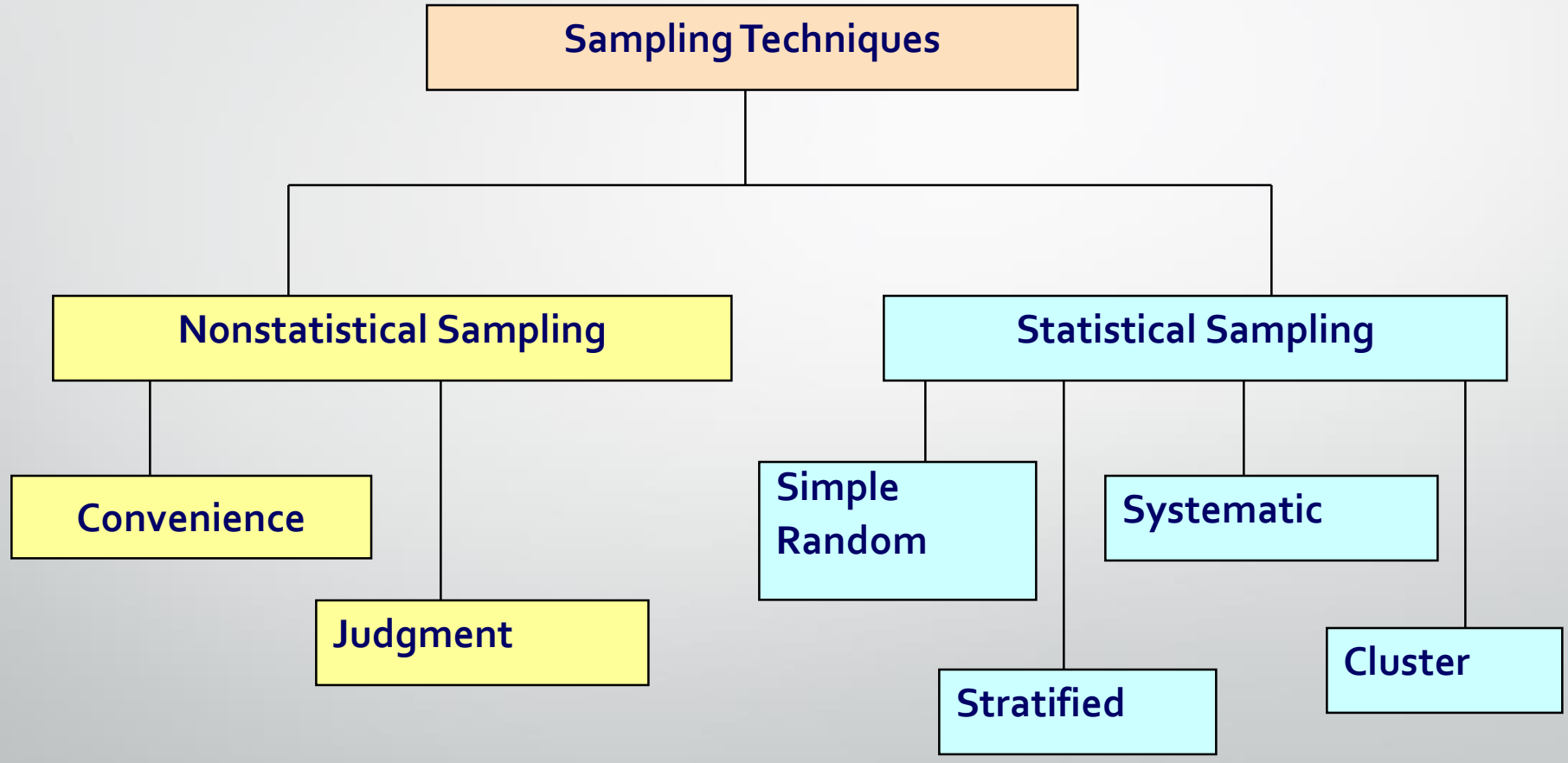
Strive for representative samples to reflect the population of interest accurately!

Sample sizes can be computed by applying the formula:

$$n = \frac{N}{1 + Ne^2}$$

where n - number of samples
N - number of population
e - margin of error

SAMPLING TECHNIQUES



Nonstatistical Sampling

- **Snowball Sampling**

  the research participants recruit other participants for a test or study. It is used when potential participants are hard to find.

- **Convenience**

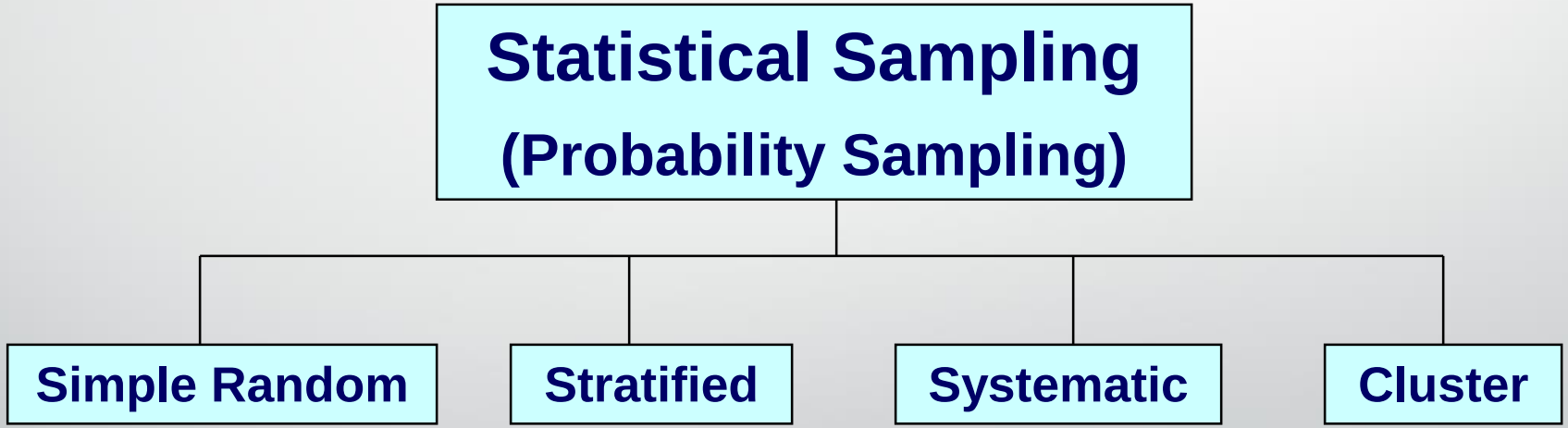
  Collected in the most convenient manner for the researcher

- **Judgmental or Purposive Sampling**

  Based on judgments about who in the population would be most likely to provide the needed information

Statistical Sampling

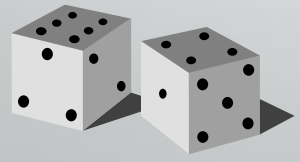
- Items of the sample are chosen based on known or calculable probabilities

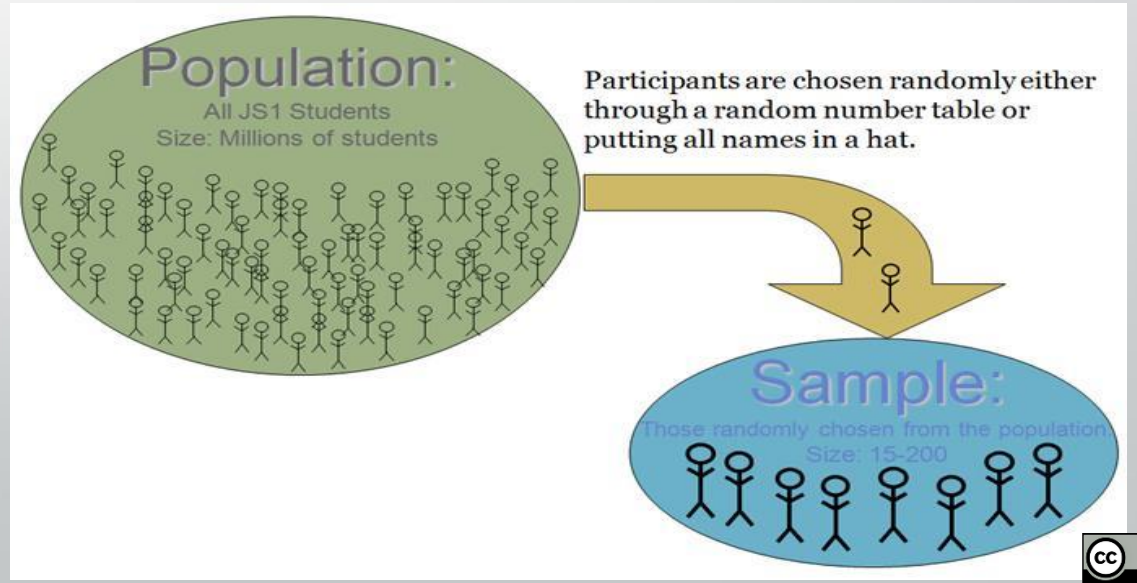


4 METHODS OF STATISTICAL SAMPLING

1. Simple Random Sampling

- Every possible sample of a given size has an equal chance of being selected
- The sample can be obtained using a table of random numbers or computer random number generator

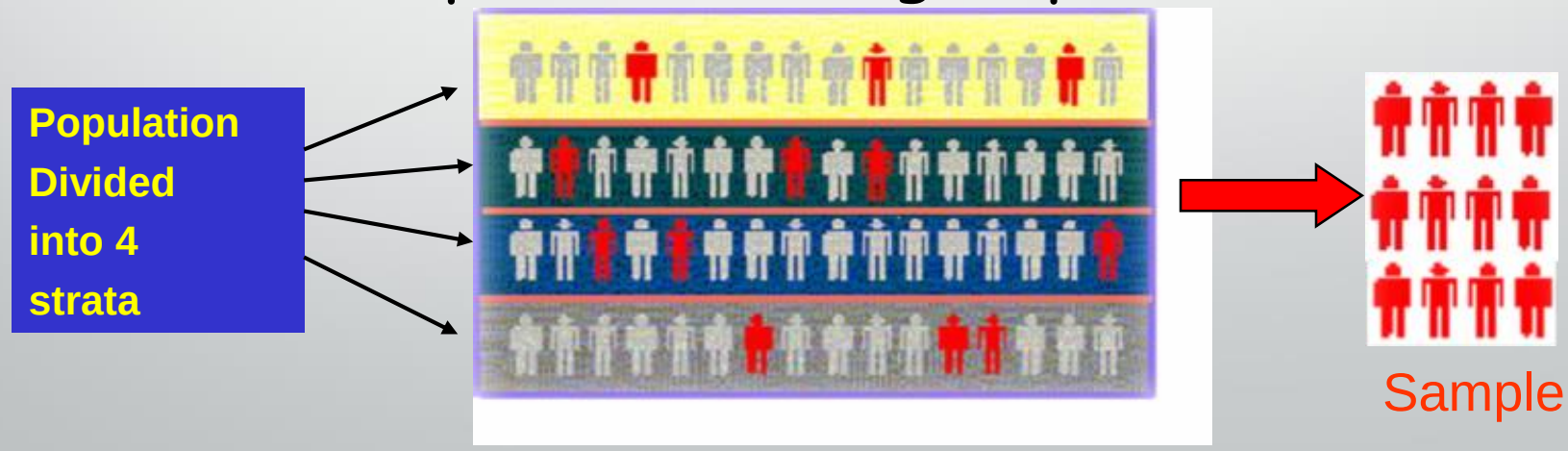




4 METHODS OF STATISTICAL SAMPLING

2. Stratified Random Sampling

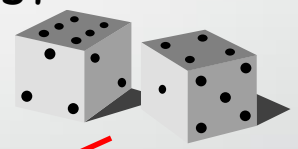
- Divide population into subgroups (called *strata*) according to some common characteristic
  - e.g., gender, income level
- Select a simple random sample from each subgroup
- Combine samples from subgroups into one



4 METHODS OF STATISTICAL SAMPLING

3. Systematic Random Sampling

- Decide on sample size: n
- Divide ordered (e.g., alphabetical) frame of N individuals into n groups of k individuals: k=N/n
- Randomly select one individual from the 1st group
- Select every kth individual thereafter

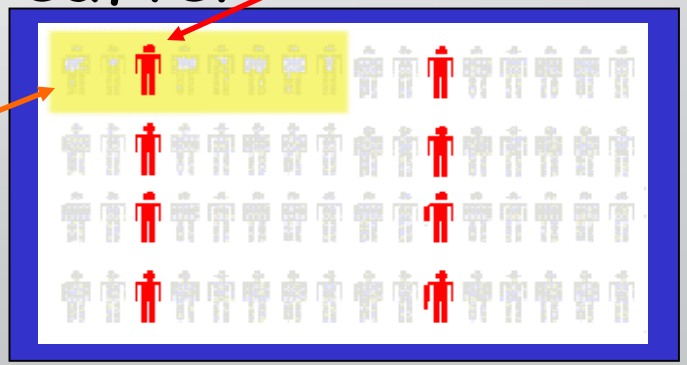


N = 64

n = 8

k = 8

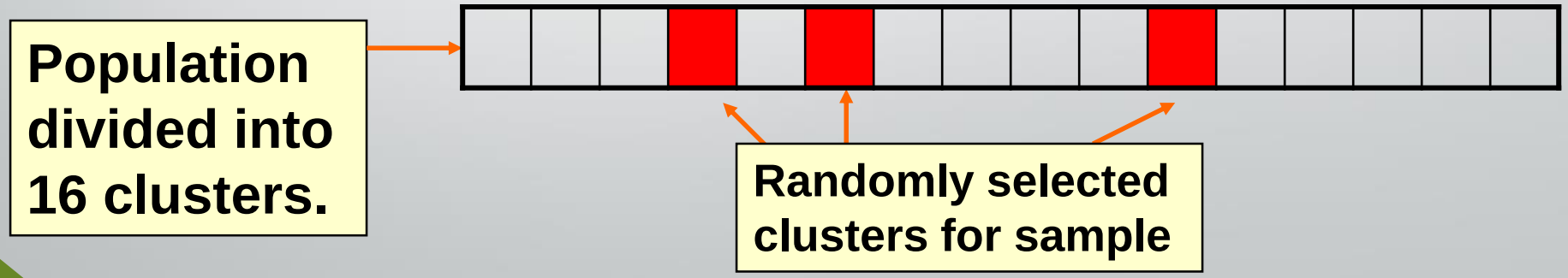
First Group



4 METHODS OF STATISTICAL SAMPLING

4. Cluster Sampling

- Divide population into several "clusters," each representative of the population (e.g., province)
- Select a simple random sample of clusters
  - All items in the selected clusters can be used, or items can be chosen from a cluster using another probability sampling technique



Key Differences

Key	Stratified Sampling	Cluster Sampling
Meaning	Stratified sampling is one, in which the population is divided into homogeneous segments, and then the sample is randomly taken from the segments.	Cluster sampling refers to a sampling method wherein the members of the population are selected at random, from naturally occurring groups called 'cluster'.
Sample	Randomly selected individuals are taken from all the strata.	All the individuals are taken from randomly selected clusters.
Selection of population elements	Individually	Collectively
Homogeneity	Within group	Between groups
Heterogeneity	Between groups	Within group
Bifurcation	Imposed by the researcher	Naturally occurring groups
Objective	To increase precision and representation.	To reduce cost and improve efficiency.

Reference: https://www.google.com/search?q=difference+between+stratified+and+cluster+sampling&tbm=isch&source=iu&ictx=1&fir=9t9tAx1AB76AIM%253A%252CaGAymGpiRZZ0DM%252C_&vet=1&usg=AI4_-kTFtBeMBJqARXICyNKruXjw98mcgA&sa=X&ved=2ahUKEwiq0M3EzPDjAhXE6XMBHReDAKgQ9QEwAHoECAAQAw#imgrc=TMdHUA7ci5gPoM:&vet=1

Classify each sample as random, systematic, stratified, or cluster

1.) In a large school district, all teachers from two buildings are interviewed to determine whether they believe the students have less homework to do now than previous years. Cluster

2.) The team needs to get a sample of 4000 students from the population and select 480 English, 1120 Science, 960 Computer Science, 840 Engineering and 600 Math students which provides a better representation of students' college majors in U.S. Stratified

3.) Every 100th hamburger manufactured is checked to determine its fat content. Systematic

4.) Mail carriers of a large city are divided into four groups according to gender (male or female) and according to whether they walk or ride on their routes. Then 10 are selected from each group and interviewed to determine whether they have been bitten by a dog last year. Stratified

PRACTICE

REFERENCE:

BLUMAN, A. (2012). *ELEMENTARY STATISTICS: A STEP BY STEP APPROACH*, 8E. MCGRAW-HILL HIGHER ED.

End of Lecture



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